Jidu Yu

Postdoctoral Researcher

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RESEARCH INTERESTS

My research develops high-fidelity computational methods for geomaterials in extreme environments, where coupled thermal, hydraulic, mechanical and chemical (THMC) processes drive large deformation and failure.

- Particle and hybrid numerical methods: MPM, DEM, MPM-DEM, MPM-FVM
- Multiphysics (THM/C) modelling of multiphase granular media, e.g., frozen soils and hydrate-bearing sediments
- Climate-driven geohazards: permafrost thaw, rainfall-induced landslides, hydrate dissociation-induced instability
- Geomechanics in extreme environments: submarine, deep underground and extraterrestrial
- Machine learning-aided multiscale and multiphysics modelling
- 3D printing of extrusion-based materials: rheology and numerical modelling

ACADEMIC APPOINTMENTS

Feb. 2026 – present
Postdoctoral Researcher
Department of Civil and Environmental Engineering, University of California, Berkeley, USA
ADVISOR Prof. Kenichi Soga

Sep. 2025 – Jan. 2026
Postdoctoral Researcher
Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology
ADVISOR Prof. Jidong Zhao

Nov. 2024 – Apr. 2025
Visiting Ph.D. Researcher
Department of Civil and Environmental Engineering, University of California, Berkeley, USA
HOST Prof. Kenichi Soga

Jul. 2019 – Oct. 2019
Visiting Student
Department of Civil, Environmental and Geomatic Engineering, University College London, UK
HOST Prof. Yi Pik Helen Cheng

EDUCATION

Aug. 2021 – Aug. 2025
Ph.D. in Civil Engineering (Geotechnical)
The Hong Kong University of Science and Technology, Hong Kong
THESIS Multiscale Modeling of Coupled Thermo-Hydro-Mechanical Behavior in Granular Media
SUPERVISOR Prof. Jidong Zhao

Sep. 2018 – Jul. 2021
M.Eng. in Hydraulic Structure Engineering
Hohai University, Nanjing, China
DISSERTATION Experimental and Theoretical Investigation of the Crushing Behavior of Granular Media
SUPERVISOR Prof. Sihong Liu

Sep. 2014 – Jul. 2018
B.Eng. in Water Conservancy and Hydropower Engineering
Hohai University, Nanjing, China
DISSERTATION Experimental Study on the Compression Behavior of Crushable Granular Materials

RESEARCH GRANTS AND PROJECTS

2026 –
Research Grants Council of Hong Kong • General Research Fund (GRF) #16210526
A multiscale digital twin for sinkhole prediction: integrating MPM, CFD-DEM, and machine learning for urban geohazard assessment
PI: Prof. Jidong Zhao Role: **Co-I** Funding: HK$1,014,504

2025 –
National Natural Science Foundation of China
A general-purpose multiscale, multiphysics computational platform for granular media
PI: Prof. Jidong Zhao Role: Participant Funding: CN¥2.3 million

2025 –
Research Grants Council of Hong Kong • Theme-based Research Scheme #T22-607/24N
A digital twin for enhancing coastal resilience against extreme storm surges in Hong Kong
Project Coordinator: Prof. Jidong Zhao Role: Participant Funding: HK$62.89 million

2023 – 2026
California Energy Commission • GFO-22-503
Performance-based monitoring and risk assessment tool for gas pipelines under natural forces
PI: Prof. Kenichi Soga Role: Participant Funding: US$4.41 million

2022 – 2024
Research Grants Council of Hong Kong • General Research Fund (GRF) #16206322
A combined physics-informed deep learning and multiscale modelling framework for simulation of thaw-induced landslide in permafrost
PI: Dr. Shiwei Zhao (Co-PI: Prof. Jidong Zhao) Role: Participant Funding: HK$802,220

2021 – 2023
Research Grants Council of Hong Kong • General Research Fund (GRF) #16211221
Multiscale modeling of thermo-hydro-mechanical (THM) behavior in granular sediments for gas hydrate recovery
PI: Prof. Jidong Zhao Role: Participant Funding: HK$911,317

HONORS AND AWARDS

2026	IJNAMG Top 10 Most-Cited Article & Top Viewed Article · Wiley	
2026	Research Travel Grant for Postdoctoral Fellow · HKUST	**HK$20,000**
2024, 2025	RedBird Academic Excellence Award · HKUST	**HK$20,000 each**
2024, 2025	Research Travel Grant for Postgraduate · HKUST	**HK$12,000 each**
2024	Overseas Research Award · HKUST	**HK$60,000**
2021 – 2025	Postgraduate Studentship · HKUST	**HK$225,120 per year**
2022	Excellent Master's Dissertation in Hydraulic Engineering · China Association of Hydraulic Engineering Education	**40 nationwide**
2022	Excellent Master's Dissertation of Jiangsu Province · Jiangsu Provincial Degree Committee	**150 province-wide**
2022	Excellent Master's Dissertation · Hohai University	
2021	RedBird PhD Scholarship Program · HKUST	**HK$40,000**
2021	Outstanding Postgraduate · Hohai University	
2021	Postgraduate "Science & Technology Star" · Hohai University	**CN¥2,000 • top 20**
2019, 2020	National Scholarship for Postgraduates · Ministry of Education of China	**CN¥20,000 each**
2018 – 2020	First Class Postgraduate Scholarship · Hohai University	**CN¥12,000 each**

2020 Postgraduate Research & Practice Innovation Program of Jiangsu Province · Jiangsu Province — **CN¥20,000**

2019 Outstanding Master Thesis Cultivation Program · Hohai University — **CN¥20,000 • one per school**

2019 First Class Prize, 9th "MathorCup" College Mathematical Modeling Challenge · Chinese Society of Optimization, Overall Planning and Economic Mathematics

2018 Second Class Prize, 15th "Huawei Cup" China Postgraduate Mathematical Contest in Modeling · China Academic Degrees & Graduate Education Association

PUBLICATIONS

Google Scholar: **337** citations · h-index **10** · i10-index **11** (Oct 2026)
First/corresponding-author papers in *JMPS* (2), *CMAME* (2), *IJNME* (1), *IJNAMG* (1), *Computers and Geotechnics* (1), *Construction and Building Materials* (1) and *JGGE* (1).
Bold: Jidu Yu * Corresponding author

Journal Articles

J20 Chen X.Y., Zhao J.D.*, **Yu J.D.** (2026). The infiltration paradox: Pore-gas pressure controls on early slope failure during rapid hydraulic loading. *Engineering Geology*, 109124. doi:10.1016/j.enggeo.2026.109124

J19 **Yu J.D.**, Zhao J.D. (2026). A virtual heat flux method for simple and accurate Neumann thermal boundary imposition in the material point method. *International Journal for Numerical Methods in Engineering*, 127(12), e70371. doi:10.1002/nme.70371

J18 Liu Y.S., Shen C.M.*, Liu S.H., **Yu J.D.**, Liu Y.C., Li B.W. (2026). Refined constitutive modelling of widely graded granular soils incorporating fractional particle breakage. *Computers and Geotechnics*, 194, 107973. doi:10.1016/j.compgeo.2026.107973

J17 Zhang C.X., Chen Y.N.*, **Yu J.D.**, Yang Z.X., Jardine R.J., Guo N. (2026). Stabilized semi-implicit material point method for hydro-mechanical coupled large deformation soil-structure interaction analyses. *International Journal for Numerical and Analytical Methods in Geomechanics*, 50(9), 3935–3954. doi:10.1002/nag.70314

J16 **Yu J.D.***, Zhao J.D.*, Liang W.J. (2026). Multiscale modeling of coupled thermo-hydro-mechanical-chemical behavior in hydrate-bearing sediment. *Journal of the Mechanics and Physics of Solids*, 210, 106512. doi:10.1016/j.jmps.2026.106512

J15 **Yu J.D.**, Zhao J.D.*, Soga K., Zhao S.W., Liang W.J. (2026). A fully coupled THMC-MPM framework for modeling phase transition and large deformation in methane hydrate-bearing sediment. *Journal of the Mechanics and Physics of Solids*, 206, 106368. doi:10.1016/j.jmps.2025.106368

J14 Liang W.J., Chandra B., **Yu J.D.**, Yin Z.Y.*, Zhao J.D. (2025). A total-Lagrangian material point method for fast and stable hydromechanical modeling of porous media. *International Journal for Numerical Methods in Engineering*, 126(19), e70135. doi:10.1002/nme.70135

J13 **Yu J.D.**, Liang W.J., Zhao J.D.* (2025). Enhancing dynamic modeling of porous media with compressible fluid: A THM material point method with improved fractional step formulation. *Computer Methods in Applied Mechanics and Engineering*, 444, 118100. doi:10.1016/j.cma.2025.118100

J12 **Yu J.D.**, Zhao J.D.*, Liang W.J., Zhao S.W. (2024). Multiscale modeling of coupled thermo-hydro-mechanical behavior in ice-bonded granular media subject to freeze-thaw cycles. *Computers and Geotechnics*, 171, 106349. doi:10.1016/j.compgeo.2024.106349

J11 **Yu J.D.**, Zhao J.D.*, Zhao S.W., Liang W.J. (2024). Thermo-hydro-mechanical coupled material point method for modeling freezing and thawing of porous media. *International Journal for Numerical and Analytical Methods in Geomechanics*, 48(13), 3308–3349. doi:10.1002/nag.3794

J10 **Yu J.D.**, Zhao J.D.*, Liang W.J., Zhao S.W. (2024). A semi-implicit material point method for coupled thermo-hydro-mechanical simulation of saturated porous media in large deformation. *Computer Methods in Applied Mechanics and Engineering*, 418, 116462. doi:10.1016/j.cma.2023.116462

J9 Shen C.M., Liu S.H.*, Wang L.J., **Yu J.D.**, Wei H., Wu P. (2022). Packing, compressibility, and crushability of rockfill materials with polydisperse particle size distributions and implications for dam engineering. *Water Science and Engineering*, 15(4), 358–366. doi:10.1016/j.wse.2022.07.003

J8 Shen C.M., **Yu J.D.**[*], Liu S.H.[*], Mao H.Y. (2021). A unified fractional breakage model for granular materials inspired by the crushing tests of dyed gypsum particles. *Construction and Building Materials*, 270, 121366. doi:10.1016/j.conbuildmat.2020.121366

J7 Shen C.M., Liu S.H.[*], **Yu J.D.**, Wang L.J. (2021). A simple scale effect model for the volumetric behavior of rockfill materials. *International Journal of Geomechanics*, 21(3), 04020266. doi:10.1061/(ASCE)GM.1943-5622.0001939

J6 **Yu J.D.**, Shen C.M.[*], Liu S.H., Cheng Y.P. (2020). Exploration of the survival probability and shape evolution of crushable particles during one-dimensional compression using dyed gypsum particles. *Journal of Geotechnical and Geoenvironmental Engineering*, 146(11), 04020121. doi:10.1061/(ASCE)GT.1943-5606.0002371

J5 Wang T., Liu S.H.[*], Feng Y., **Yu J.D.** (2018). Compaction characteristics and minimum void ratio prediction model for gap-graded soil-rock mixture. *Applied Sciences*, 8(12), 2584. doi:10.3390/app8122584

Journal Articles in Chinese

J4 **Yu J.D.**, Liu S.H.[*], Shen C.M.[*], Mao H.Y. (2022). Breakage and shape evolution of dyed gypsum particles under one-dimensional compression. *Journal of Hohai University (Natural Sciences) (*河海大学学报（自然科学版）*)*, 50(1), 110–116. (In Chinese) doi:10.3876/j.issn.1000-1980.2022.01.016

J3 Wei H., Shen C.M., Liu S.H., **Yu J.D.** (2020). Experimental study on compression and crushing characteristics of coarse granular materials considering influence of gradations. *Journal of Hohai University (Natural Sciences) (*河海大学学报（自然科学版）*)*, 48(2), 182–188. (In Chinese) doi:10.3876/j.issn.1000-1980.2020.02.014

J2 **Yu J.D.**, Shen C.M.[*], Liu S.H. (2020). Breakage and migration of dyed gypsum particles under one-dimensional compression. *Chinese Journal of Rock Mechanics and Engineering (*岩石力学与工程学报*)*, 39(5), 1071–1079. (In Chinese) doi:10.13722/j.cnki.jrme.2019.0739

J1 **Yu J.D.**, Liu S.H.[*], Wang T., Wei H. (2019). Experimental research on compaction characteristics of gap-graded coarse-grained soils. *Chinese Journal of Geotechnical Engineering (*岩土工程学报*)*, 41(11), 2142–2148. (In Chinese) doi:10.11779/CJGE201911021

Manuscripts under Review

R2 Liu X.Y., Gao G., **Yu J.D.**, Meguid M.A., Zhang F.S. (2026). Micromechanical characterization of fracture evolution in confined jointed rocks using a 3D grain-based model. Submitted to *Rock Mechanics and Rock Engineering*.

R1 **Yu J.D.**, Chandra B.[*], Wilkes C., Zhao J.D., Soga K. (2025). An elasto-viscoplastic thixotropic model for fresh concrete capturing flow-rest transition. Submitted to *Cement and Concrete Composites*. arXiv:2512.23364. arXiv:2512.23364

Conference Papers and Abstracts

C7 **Yu J.D.**, Zhao J.D.[*] (2026). High-fidelity multiscale framework for modelling permafrost thaw-related geohazards. *Proceedings of the 21st International Conference on Soil Mechanics and Geotechnical Engineering (ICSMGE)*, 1743–1746, Vienna, Austria.

C6 **Yu J.D.**, Zhao J.D.[*] (2025). A multiphysics-multiscale MPM-DEM framework for modeling climate-driven geohazards in phase-change granular media. *IX International Conference on Particle-based Methods (Particles 2025)*, 20–22 October 2025, Barcelona, Spain.

C5 **Yu J.D.**, Zhao J.D.[*] (2025). Hybrid explicit-implicit THMC-MPM for modeling phase transition and large deformation in hydrate-bearing sediment. *The 2nd Workshop on Material Point Method & Extreme Deformation Problems (MPM 2025)*, 12–14 September 2025, Xi'an, China.

C4 **Yu J.D.**, Zhao J.D.[*], Zhao S.W., Liang W.J. (2024). Thermo-hydro-mechanical coupled material point method for simulating the freeze-thaw behavior of porous media. *16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics (WCCM-PANACM 2024)*, July 2024, Vancouver, Canada.

C3 **Yu J.D.**, Zhao J.D.[*], Zhao S.W., Liang W.J. (2024). Multiphysics simulation of freezing and thawing granular media using material point method. *GeoShanghai International Conference 2024*, May 2024, Shanghai, China. *IOP Conference Series: Earth and Environmental Science*, 1330, 012035. doi:10.1088/1755-1315/1330/1/012035

C2 **Yu J.D.**, Liang W.J., Zhao S.W., Zhao J.D.[*] (2023). A semi-implicit MPM for modeling the thermal effect in saturated porous media. *International Symposium on Innovations in Geotechnical Engineering towards Sustainability (IGES 2023)*, Hong Kong, China.

C1 **Yu J.D.**, Zhao J.D.*, Zhao S.W., Liang W.J. (2023). An explicit-implicit THM-coupled MPM for saturated porous media. *The 26th Annual Conference of HKSTAM 2023 in conjunction with the 18th Shanghai–Hong Kong Forum on Mechanics and Its Application*, Hong Kong, China, p. 30.

Thesis

T1 **Yu J.D.** (2025). Multiscale modeling of coupled thermo-hydro-mechanical behavior in granular media. Ph.D. thesis, The Hong Kong University of Science and Technology, Hong Kong. doi:10.14711/thesis-hdl152441

INVITED TALKS

Nov. 2026 **Phase transition meets large deformation: Multiphysics and multiscale modeling of frozen soils and hydrate-bearing sediments**
MechBytes Seminar, USACM Student Chapter, University of California, Berkeley, CA, USA

Jan. 2026 **Multiphysics and multiscale modeling of geomaterials in large deformation: taking frozen soil and hydrate soil as examples**
College of Architecture and Civil Engineering, Beijing University of Technology, Beijing, China

2024 **Multiscale modeling the thermo-hydro-mechanical behavior of frozen soils**
College of Water Conservancy and Hydropower Engineering, Hohai University, Nanjing, China

TEACHING EXPERIENCE

Teaching Assistant, The Hong Kong University of Science and Technology

2023 – 2025 **Undergraduate Final-Year Projects (2023–24, 2024–25)**

Fall 2023 **CIVL 4750 Numerical Solutions to Geotechnical Problems**

Spring 2023 **CIVL 3740 Geotechnical Analysis and Design**

Spring 2022 **CIVL 4320 Structural Steel Design**

PROFESSIONAL SERVICE

2026 – **Ad hoc reviewer**, U.S. National Science Foundation (NSF)

2025 – **Journal reviewer**
Advances in Engineering Software; Ain Shams Engineering Journal; Chemical Engineering Science; Computational Particle Mechanics; Computer Methods in Applied Mechanics and Engineering; Computers and Geotechnics; Construction and Building Materials; Energy Technology; Engineering Geology; Finite Elements in Analysis and Design; Geoderma; International Journal for Numerical Methods in Fluids; International Journal of Geomechanics; International Journal of Heat and Mass Transfer; International Journal of Mining Science and Technology; Journal of Computational Physics; Journal of Rock Mechanics and Geotechnical Engineering; Journal of Traffic and Transportation Engineering (English Edition); Measurement; Mechanics of Materials; Physics and Chemistry of the Earth, Parts A/B/C; Powder Technology; Simulation Modelling Practice and Theory; Thin-Walled Structures; Transportation Geotechnics; Tunnelling and Underground Space Technology